New bounds for small two- and four-colour Ramsey numbers

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Abstract

We give upper bounds for the Ramsey numbers $\mathrm{R}(3,k)$ with $24 \le k \le 50$, for which Radziszowski's survey lists no upper bounds. They come from an exact bootstrap of lower bounds on the number of edges of triangle-free graphs with bounded independence number. It starts from the published values and lower bounds of Goedgebeur and Radziszowski. For instance, $\mathrm{R}(3,24) \le 201$, $\mathrm{R}(3,30) \le 297$ and $\mathrm{R}(3,50) \le 729$. If the table of Backelin's unrefereed report is also used, the bounds are 1 to 3 lower for $24 \le k \le 36$, except $k = 35$.

Deleting vertices from the quartic residue graphs of Lindsay and Cain gives $\mathrm{R}(4,21) \ge 260$ and $\mathrm{R}(4,24) \ge 370$. Finally, we use the four-colour form of the neighbourhood recurrence, with $\mathrm{R}(3,4,4) \le 73$ and $\mathrm{R}(4,4,4) \le 198$ from a companion paper and the survey's $\mathrm{R}(3,3,3,4) \le 149$. This gives $\mathrm{R}(3,3,4,4) \le 442$, $\mathrm{R}(3,4,4,4) \le 1521$ and $\mathrm{R}(4,4,4,4) \le 6082$. All certificates and witnesses are checked exactly by programs in the ancillary files.

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1 Introduction

For integers $s,t \ge 1$, the Ramsey number $\mathrm{R}(s,t)$ is the least n such that every red-blue colouring of the edges of K_n contains a red K_s or a blue K_t; equivalently, every graph on n vertices contains a K_s or an independent set of size t. Multicolour numbers $\mathrm{R}(t_1,\dots,t_r)$ are defined in the same way. The current state of knowledge is collected in Radziszowski's dynamic survey [15], whose revision 18 we use for all comparisons, together with the later preprint [19].

This paper is a companion to a study of three-colour numbers [5], in which local Ramsey bounds inside colour neighbourhoods are combined with double counting and with flag-algebra certificates. The present paper gives new bounds for two-colour numbers. The upper bounds are obtained by adapting the ladder procedure of [5, Section 5], and the lower bounds by deleting vertices from quartic residue graphs. It also derives consequences for four-colour numbers. Our results are listed in Table 1.

The upper bounds for $\mathrm{R}(3,k)$ (Section 2) and the lower bounds for $\mathrm{R}(4,k)$ (Section 3) are computer-assisted. Floating-point solvers and local search were used only to find certificates

number	previous	new	where
$R(3,k)$, $24 \le k \le 50$	no upper bound [15]	Table 2	Theorem 2.1
$R(4,21)$	≥ 242 [18]; ≥ 257 ([19] and [15, 2.3(e)])	≥ 260	Theorem 3.1
$R(4,24)$	≥ 343 (implicit in [14])	≥ 370	Theorem 3.1
$R(3,3,4,4)$	≤ 450 [15]	≤ 442	Theorem 4.1
$R(3,4,4,4)$	≤ 1576 [15] (implicit in [3])	≤ 1521	Theorem 4.1
$R(4,4,4,4)$	≤ 6306 [15]; ≤ 6301 (implicit in [3])	≤ 6082	Theorem 4.1

Table 1: Our results. For $R(3,k)$ with $k \ge 24$, iterating $R(3,k) \le R(3,k-1)+k$, which is strict when both terms are even, or Shearer's bound [17], gives weaker explicit bounds (Table 2). Where the two rows of Table 2 differ, the smaller bound also uses the unrefereed report [2].

and witnesses, and in the negative experiments that we report. Every certificate and witness is checked in exact arithmetic by a program that does not share code with the search (Section 5).

The four-colour bounds (Section 4) follow by hand from $R(3,3,3,4) \le 149$ [15] and from $R(3,4,4) \le 73$ and $R(4,4,4) \le 198$, which are proved in [5, Theorems 8.3 and 8.2] by certificates checked in exact arithmetic by computer.

2 Upper bounds for $R(3,k)$

A $(3,k;n)$-graph is a triangle-free graph on n vertices with no independent set of size k. A $(3,k;n,e)$-graph is one with e edges, and $e(3,k,n)$ is the least such e. By analogy with [5, Section 3], where the term is used for three-colourings, $m_k(n)$ is a *valid lower bound* if every $(3,k;n)$-graph has at least $m_k(n)$ edges.

For a vertex v of a $(3,k;n,e)$-graph G, $N(v)$ is independent, so $d_v \le k-1$. Moreover, $G_v = G - N[v]$, where $N[v] = N(v) \cup \{v\}$, is a $(3,k-1;n-1-d_v)$-graph, as an independent set of G_v extends by v. As no edge lies inside $N(v)$, G_v has $e - Z_v$ edges, where

$$Z_v = \sum_{u \in N(v)} d_u.$$

As $\sum_v Z_v = \sum_v d_v^2$, valid lower bounds m_{k-1} give

$$\sum_v \big(d_v^2 + m_{k-1}(n-1-d_v)\big) \le ne.$$

This is the inequality of Graver and Yackel [9], on which Goedgebeur and Radziszowski [7] base their bounds for $k \ge 11$. It is the two-colour analogue of the density inequality [5, Lemma 3.3]. As $ne = \sum_v nd_v/2$, it says

$$\sum_v \varphi(d_v) \le 0 \qquad \text{where} \qquad \varphi(d) = d^2 + m_{k-1}(n-1-d) - nd/2.$$

The bootstrap. We bootstrap as in the ladder of [5, Section 5], with this inequality in place of the density inequality. Level $k \ge 11$ uses the bound B_{k-1} on $R(3,k-1)$ and the valid lower bounds m_{k-1} from level $k-1$. Level 11 starts from $B_{10} = 42$ [7, Theorem 6] and the values and lower bounds for $e(3,10,n)$ in [7].

Level k treats the orders $n = 0, 1, 2, \ldots$ in turn, records a valid lower bound $m_k(n)$, and stops at the first order it excludes, B_k, that is, the first n at which it shows that no $(3, k; n)$-graph exists. So $\mathrm{R}(3, k) \le B_k$. In the same way, a value e, or a degree sequence, is excluded at order n when no $(3, k; n, e)$-graph exists (with that degree sequence). We run the procedure twice, giving two *chains*: one from the inputs of [7] alone, and one that also uses the lower bounds of [2] described after the list below. At order n the procedure is as follows.

(i) The admissible degrees are the $d \le \min(k-1, n-1)$ with $n - 1 - d \le B_{k-1} - 1$.

(ii) An order $n \ge 1$ is excluded if $\varphi(d) > 0$ at every admissible d (in particular if there is none), as $\sum_v \varphi(d_v)$ would then be a sum of n positive terms.

(iii) Otherwise let e_0 be the largest of the following bounds. The first is the bound of Radziszowski and Kreher [7, Theorem 2]. With $K = k - 1$, it gives the exact values

$$e(3,k,n) = \begin{cases} 0 & \text{for } n \le K, \\ n - K & \text{for } K < n \le 2K, \\ 3n - 5K & \text{for } 2K < n \le 5K/2, \\ 5n - 10K & \text{for } 5K/2 < n \le 3K, \\ 6n - 13K & \text{for } 3K < n \le 13K/4 - 1, \end{cases}$$

and $e(3, k, n) \ge 6n - 13K$ for all n. The second exists when φ is positive at the least admissible d but not at every admissible d. Let $(d', \varphi(d'))$ and $(d'', \varphi(d''))$, with $d' < d''$, be consecutive vertices of the lower convex hull of the points $(d, \varphi(d))$, d admissible, with $\varphi(d') > 0 \ge \varphi(d'')$, and let $\alpha - \beta d$ be the line through them. The second bound is $\lceil n\alpha/(2\beta) \rceil$. This line has $\beta > 0$ and lies on or below φ at every admissible d, so

$$0 \ge \sum_v \varphi(d_v) \ge n\alpha - 2\beta e.$$

(iv) The extra step below, at the orders where the search applied it (each is recorded in the certificates), may exclude e_0 and some larger values of e. The recorded $m_k(n)$ is the least $e \ge e_0$ that it does not exclude. If it excludes every e from e_0 to $\lfloor n(k-1)/2 \rfloor$, the most edges possible when every degree is at most $k - 1$, then n is excluded.

In the chain with [2], the lower bounds of [2] on $e(3, k, n)$ for $11 \le k \le 13$ and $35 \le n \le 43$ are a further candidate for e_0 in (iii); at $k = 10$ they agree with those of [7]. Deleting a vertex as in [5, Lemma 3.5(a)] gives $m_k(n-1) + \max\{0, n - B_{k-1}\}$, as every degree is at least $n - B_{k-1}$. This never exceeds e_0 in either chain, so the certificates do not use it.

An extra step. We add one step, for $e = e_0, e_0 + 1, \ldots$ in turn. We enumerate every degree sequence with sum $2e$ and admissible entries, each at most $e - m_k(n-1)$, that satisfies the Graver–Yackel inequality. The cap on the entries holds as deleting v leaves a $(3, k; n-1)$-graph with $e - d_v$ edges.

We then try to exclude each sequence by a small linear programme. The *class* of an entry a is the set of the c_a vertices of degree a. The variables are the numbers $x_{ab} = x_{ba}$ of edges between the classes of $a \ne b$ and the numbers x_{aa} of edges inside a class. Counting edge ends, write $x_{a \to b} = x_{ab}$ for $a \ne b$ and $x_{a \to a} = 2x_{aa}$, so that $\sum_{d_v = a} Z_v = \sum_b b\, x_{a \to b}$. With $z_a = e - m_{k-1}(n - 1 - a)$, every $(3, k; n, e)$-graph with that degree sequence satisfies, for all entries a and b:

- $\sum_b x_{a\to b} = a\,c_a$;
- $\sum_b b\,x_{a\to b} \le c_a z_a$, as G_v has $e - Z_v \ge m_{k-1}(n-1-a)$ edges;
- $\sum_b b\,x_{a\to b} \ge c_a\big(e - \lfloor (n-1-a)(k-2)/2 \rfloor\big)$, as G_v is a $(3, k-1)$-graph and so has maximum degree at most $k-2$;
- $x_{ab} \le c_a c_b$ for $a \ne b$;
- $\sum x_{ab} \le \lfloor N_S^2/4 \rfloor$, summed over $a \le b$ in S, for every set S of classes with N_S vertices in all (Mantel's theorem);
- $\sum_{b' \le b} x_{a\to b'} \ge t\,c_a$, with $t = t(a,b)$ as follows.

For the last constraint, let q be the least entry above b, and let v be a vertex of degree a. Remove from the sequence one term a, that of v. For $0 \le j \le a$, let $L(j)$ be the sum of the j smallest remaining terms at most b, counted with multiplicity, plus $(a-j)q$. Put $L(j) = \infty$ if fewer than j such terms remain, or if no entry of the sequence exceeds b (so that q is undefined) and $j < a$. If v has exactly j neighbours of degree at most b, then $L(j) \le Z_v \le z_a$ (so no such v exists when $L(j) = \infty$). So v has at least t such neighbours, where t is the least j with $L(j) \le z_a$, and summing over the class of a gives the last constraint. If there is no such j, no vertex of degree a exists and the sequence is excluded by this *neighbour-count test*.

If every sequence is excluded, so is e, and the step moves on to $e+1$. Each exclusion is certified either by the neighbour-count test or by rational Farkas multipliers, which combine these constraints, with $x \ge 0$, into an inequality $0 \ge c$ with $c > 0$.

Theorem 2.1 (computer-assisted)**.** *From the published values and lower bounds of [7, Theorem 2 and Tables 3–5] and* $\mathrm{R}(3,10) \le 42$ *[7, Theorem 6],* $\mathrm{R}(3,k)$ *is at most the value in the row "[7] only" of Table 2 for* $24 \le k \le 50$*. If the lower bounds on* $e(3,k,n)$ *in the table of [2, Section 5] are also used,* $\mathrm{R}(3,k)$ *is at most the value in the row "with [2]".*

k	24	25	26	27	28	29	30	31	32
[7] only	201	216	231	247	264	280	297	315	333
with [2]	200	215	230	245	261	278	295	312	331
k	33	34	35	36	37	38	39	40	41
[7] only	351	370	389	409	429	450	471	492	513
with [2]	349	369	389	408	429	450	471	492	513
k	42	43	44	45	46	47	48	49	50
both	536	559	582	606	630	654	679	704	729

Table 2: Upper bounds for $\mathrm{R}(3,k)$ from Theorem 2.1. The two rows coincide for $k = 35$ and $k \ge 37$, and for $k \ge 42$ they are shown as one, "both". The survey [15] lists no upper bounds for these k. For comparison, $\mathrm{R}(3,k) \le \mathrm{R}(3,k-1) + k$, strict when both terms are even, iterated from the survey's $\mathrm{R}(3,23) \le 185$, gives 209, 234, 259, 371 and 1171 for $k = 24, 25, 26, 30, 50$. Shearer's bound [17] $\mathrm{R}(3,k) \le \lfloor (k-1)/f(k-1) \rfloor + 1$, with $f(d) = (d\ln d - d + 1)/(d-1)^2$, gives 223 for $k = 24$ and 792 for $k = 50$.

The two chains. Above level 10, each chain uses only the bounds B_{k-1} that it proves itself, never the unpublished values of [15]. Neither chain needs the bound $\mathrm{R}(3,10) \le 41$ of Angeltveit [1]. The report [2] is a preprint that gives little of its proofs, so the second row rests on inputs without published proofs. The first row is 1 to 3 larger for $24 \le k \le 34$ and $k = 36$, and equals the second for $k = 35$ and $k \ge 37$.

Without the extra step, the chain from [7] alone gives the same bounds for $11 \le k \le 50$, except $\mathrm{R}(3,20) \le 146$ and $\mathrm{R}(3,35) \le 390$. This run is not among the certificates in the ancillary files. The chain from [7] alone also reaches $e(3,13,51) \ge 178$ and $e(3,14,67) \ge 335$, which the comments column of [7, Tables 9 and 10] credits to Backelin. It raises seven further lower bounds of [7, Tables 9–11] by one, to

$$e(3,13,58) \ge 261, \qquad e(3,14,69) \ge 366, \qquad e(3,14,70) \ge 382,$$
$$e(3,15,n) \ge 516,\ 534,\ 553 \text{ and } 573 \quad \text{for } n = 82,\dots,85.$$

These are values $m_k(n)$ recorded in the certificate of that chain, so the checker of Theorem 2.1 verifies them. They lie beyond the range $n \le 43$ of [2]. Backelin's unpublished manuscript, to which [15] credits $\mathrm{R}(3,17) \le 109$, may already contain them.

Checking the certificates. The certificates are checked by a program that does not import the generating code. It enumerates the degree sequences itself, rebuilds each constraint from its definition, and verifies the multipliers in exact arithmetic.

- The chain with [2] has one row for each level $11 \le k \le 50$ and each order $0 \le n < B_k$, 13,376 rows in all. Each row records $m_k(n)$ and the bound of (iii) that gives e_0: the Radziszowski–Kreher bound, a value of [2], or the two ends of the hull edge. At the orders where the extra step is used, it also records a certificate for every degree sequence at each excluded e. A further row per level, not counted here, certifies the exclusion of B_k: by (ii) at every level except $k = 16$, where the extra step excludes every e at $n = 97$. The chain excludes 765 degree sequences at 53 triples (k,n,e) (including those at $k = 16$, $n = 97$), 686 by the neighbour-count test and 79 by Farkas multipliers.
- The chain from [7] alone has 13,403 rows and excludes 766 sequences at 50 triples, 684 and 82 respectively. All 40 exclusions of B_k in it are by (ii).

Comparison with the survey. For $11 \le k \le 23$ the two chains give the following bounds B_k, against the upper bounds that [15] lists.

k	11	12	13	14	15	16	17	18	19	20	21	22	23
survey [15]	50	59	68	77	87	97	109	120	132	145	157	171	185
[7] only	50	59	68	77	87	98	109	121	133	145	159	173	187
with [2]	50	59	68	77	87	97	109	121	132	145	158	172	185

So for $11 \le k \le 23$ the procedure improves no value of [15]. From [7] alone it gives $\mathrm{R}(3,17) \le 109$ and $\mathrm{R}(3,20) \le 145$, which [15] credits to an unpublished manuscript of Backelin and to Lesser's thesis [12], respectively. With the table of [2] it gives $\mathrm{R}(3,16) \le 97$ and $\mathrm{R}(3,19) \le 132$, the values that [15] lists, citing [2].

With the table of [2] it also gives $\mathrm{R}(3,23) \le 185$. The survey [15] credits this bound to a personal communication of Backelin; it now has a derivation that can be checked, given the

inputs of [2]. At $k = 18$, 21 and 22 the values of [15] come from unpublished sources, and the chain with [2] misses each by one.

For $k = 18$, both chains give $m_{18}(120) = 1020 = 120 \cdot 17/2$ from the hull. As every degree is at most 17, a $(3, 18; 120)$-graph would have exactly 1020 edges and be 17-regular. Then every G_v would be a $(3, 17; 102, 731)$-graph, as $120 - 1 - 17 = 102$ and $1020 - 17^2 = 731$. So $e(3, 17, 102) \geq 732$, one more than our $m_{17}(102) = 731$, would exclude $n = 120$ and give $\mathrm{R}(3, 18) \leq 120$.

3 Lower bounds for $\mathrm{R}(4, k)$

For a prime $p \equiv 1 \pmod 8$ let Q_p be the circulant graph on $\mathbb{Z}_p$ in which $x \sim y$ if and only if $x - y$ is a nonzero fourth power. The relation is symmetric, as -1 is a fourth power. Lindsay and Cain [14, Proposition 2.1] computed with cliquer that Q_{313} and Q_{457} have clique number 3 and independence numbers

$$\alpha(Q_{313}) = 21 \qquad \text{and} \qquad \alpha(Q_{457}) = 24.$$

The ancillary checks confirm $\alpha(Q_{313}) = 21$ and show that Q_{457} has an independent 24-set.

Theorem 3.1 does not use the upper bounds on $\alpha(Q_p)$. An independent set of size at least k contains one of size k. So if a vertex set S meets every independent k-set of Q_p, then $Q_p - S$ is K_4-free and has no independent k-set. Step 3 below checks this for $(p, k) = (313, 21)$ and $(457, 24)$, and the K_4-freeness of both graphs $Q_p - S$ is also checked directly.

Theorem 3.1 (witnesses checked by computer)**.** $\mathrm{R}(4, 21) \geq 260$ *and* $\mathrm{R}(4, 24) \geq 370$.

Proof. Step 1: symmetries and cosets. The maps $x \mapsto ax + b$ with a a nonzero fourth power are automorphisms of Q_p. Let O_1, O_2, O_3 be the cosets of the group of nonzero fourth powers in $\mathbb{Z}_p^*$ other than the group itself, in a fixed order, and let r_j be the least element of O_j.

Step 2: representatives. Let $k \geq 2$, let J be an independent k-set and let $u \in J$. The translate $J - u = \{x - u : x \in J\}$ is an independent k-set that contains 0, so its other elements are not fourth powers and lie in $O_1 \cup O_2 \cup O_3$. Let j be least with $J - u$ meeting O_j, say at w. Then r_j/w is a fourth power, and multiplying by it maps $J - u$ to an independent k-set that contains 0 and r_j and still avoids $O_1, \ldots, O_{j-1}$, as multiplication by a fourth power fixes each coset. Call an independent k-set a *representative* if, for some j, it contains 0 and r_j and avoids $O_1, \ldots, O_{j-1}$. So every independent k-set is the image of a representative under one of the maps above.

Step 3: the computation. The sets S, of 54 and 88 vertices, were found by a weighted local search and are listed in the ancillary files. An exact program lists all representatives for $(p, k) = (313, 21)$ and $(457, 24)$, by an exhaustive search for independent $(k-2)$-sets in the common non-neighbourhood of 0 and r_j outside $O_1, \ldots, O_{j-1}$. It then checks that S meets every image of every representative. So $Q_{313} - S$ is a K_4-free graph on 259 vertices with no independent 21-set, and $Q_{457} - S$ one on 369 vertices with no independent 24-set. □

Verification. Both graphs are rebuilt from the lists of deleted vertices by programs that share no code with the search. The 259-vertex graph is also checked by a direct exhaustive

search for K_4 and for independent 21-sets, and the 369-vertex graph by one for K_4 and for independent 24-sets. Neither search uses symmetry, so each is independent of the orbit argument above. □

Comparison with earlier bounds. The previous bounds are as follows.

- For $\mathrm{R}(4,21)$, the survey has 242 [18], and $\mathrm{R}(4,20) \geq 252$ from a recent preprint [19] gives 257 by the inequality $\mathrm{R}(4,l) \geq \mathrm{R}(4,l-1)+5$ [15, 2.3(e)].
- For $\mathrm{R}(4,24)$, the non-neighbourhood of a vertex of Q_{457} (Q_{457} is 114-regular) has 342 vertices, is K_4-free and has no independent 24-set, as such a set would extend by the vertex. This gives $\mathrm{R}(4,24) \geq 343$. The survey lists no value for $\mathrm{R}(4,24)$. The inequality [15, 2.3(e)], applied twice to $\mathrm{R}(4,22) \geq 314$ [14], gives 324.

The method does not reach $\mathrm{R}(4,23) \geq 320$, which would improve the bound 319 that this inequality gives from $\mathrm{R}(4,22) \geq 314$. That needs a set of at most $457-319=138$ vertices meeting every independent 23-set of Q_{457}, and the smallest we found has 147.

4 Four colours

For $\mathbf{t} = (t_1,\ldots,t_4)$ and the unit vectors $\mathbf{e}_i$, the four-colour form of the neighbourhood recurrence [15, 6.1(a)] is

$$\mathrm{R}(\mathbf{t}) \leq \sum_{i=1}^{4} \mathrm{R}(\mathbf{t}-\mathbf{e}_i) - 2.$$

Indeed, consider a colouring of K_n with no colour-j K_{t_j} for any j. The colour-i neighbourhood $N_i(v)$ of a vertex v, whose size is the colour-i degree of v, contains no colour-i K_{t_i-1}, which would extend by v, and no colour-j K_{t_j} for $j \neq i$. So it has at most $\mathrm{R}(\mathbf{t}-\mathbf{e}_i)-1$ vertices. Summing over i gives

$$n-1 \leq \sum_i (\mathrm{R}(\mathbf{t}-\mathbf{e}_i)-1).$$

So such a colouring has $n \leq \sum_i \mathrm{R}(\mathbf{t}-\mathbf{e}_i) - 3$, which is the recurrence.

The recurrence improves to $\mathrm{R}(\mathbf{t}) \leq \sum_i \mathrm{R}(\mathbf{t}-\mathbf{e}_i) - 3$ when $\sum_i \mathrm{R}(\mathbf{t}-\mathbf{e}_i)$ is even and some $\mathrm{R}(\mathbf{t}-\mathbf{e}_i)$ is even. Indeed, take a colouring as above of K_n with $n = \sum_i \mathrm{R}(\mathbf{t}-\mathbf{e}_i) - 3$, which is odd. Then the display holds with equality, so every vertex has colour-i degree exactly $\mathrm{R}(\mathbf{t}-\mathbf{e}_i)-1$ for each i. For an i with $\mathrm{R}(\mathbf{t}-\mathbf{e}_i)$ even this degree is odd, and a graph on an odd number of vertices cannot have all its degrees odd.

Both statements hold with the same proofs when each $\mathrm{R}(\mathbf{t}-\mathbf{e}_i)$ is replaced by a proven upper bound for it. An entry equal to 2 may be dropped, leaving a number with fewer colours, as a colouring with no colour-i K_2 has no colour-i edges.

Theorem 4.1 (from computer-assisted inputs)**.** $\mathrm{R}(3,3,4,4) \leq 442$, $\mathrm{R}(3,4,4,4) \leq 1521$ *and* $\mathrm{R}(4,4,4,4) \leq 6082$.

Proof. With $\mathrm{R}(3,4,4) \leq 73$ [5, Theorem 8.3] and $\mathrm{R}(3,3,3,4) \leq 149$ [15], the recurrence gives

$$\mathrm{R}(3,3,4,4) \leq 2\cdot 73 + 2\cdot 149 - 2 = 442.$$

Parity does not apply, as every term is odd.

With $\mathrm{R}(4,4,4) \le 198$ [5, Theorem 8.2] it then gives

$$\mathrm{R}(3,4,4,4) \le 198 + 3 \cdot 442 - 2 = 1522.$$

The sum $198 + 3 \cdot 442 = 1524$ is even, as are all four terms, so parity gives $\mathrm{R}(3,4,4,4) \le 1524 - 3 = 1521$.

Finally,

$$\mathrm{R}(4,4,4,4) \le 4 \cdot 1521 - 2 = 6082,$$

and parity does not apply, as 1521 is odd. □

The inputs and earlier bounds. The two inputs from [5] are computer-assisted. The survey's $\mathrm{R}(3,3,3,4) \le 149$ is itself the recurrence with parity, from $\mathrm{R}(3,3,4) = 30$ [4] and $\mathrm{R}(3,3,3,3) \le 62$ [6]: as $3 \cdot 30 + 62 = 152$ and 30 are even, it gives $152 - 3 = 149$. From the survey's $\mathrm{R}(3,4,4,4) \le 1576$, the recurrence with parity gives $\mathrm{R}(4,4,4,4) \le 4 \cdot 1576 - 3 = 6301$, as $4 \cdot 1576$ and all four terms are even. The survey lists 6306 and notes that 6301 is implicit in [3].

With the survey's $\mathrm{R}(3,3,4,4) \le 450$ in place of 442, that is, without $\mathrm{R}(3,4,4) \le 73$, the bound $\mathrm{R}(4,4,4) \le 198$ alone gives $\mathrm{R}(3,4,4,4) \le 1545$ and $\mathrm{R}(4,4,4,4) \le 6178$. The survey's $\mathrm{R}(3,4,4,4) \le 1576$, which it marks as implicit in [3], comes from Boza's analogue of the parity clause [3, Theorem 2.1], which counts the triangles of one colour. It lowers the right-hand side $P = \sum_i \mathrm{R}(\mathbf{t} - \mathbf{e}_i) - 2$ of the recurrence by one when three conditions modulo 3 all hold, the first being $P \not\equiv 1 \pmod 3$. From Boza's $\mathrm{R}(4,4,4) \le 229$ and the survey's $\mathrm{R}(3,3,4,4) \le 450$ it gives $229 + 3 \cdot 450 - 2 - 1 = 1576$. With our bounds it gives nothing more. At the three steps P is 442, 1522 and 6082 (before any parity improvement), all $\equiv 1 \pmod 3$, so the first condition fails.

Four triangle-free colours. We also tried to improve $\mathrm{R}(3,3,3,3) \le 62$ [6], that is, to exclude colourings of K_{61} with four triangle-free colours, by certificates analogous to those for $\mathrm{R}(4,4,4)$ in [5, Sections 7 and 8]. Such a certificate exists only if a flag relaxation, here with four- or five-vertex patterns and with four-colour analogues of the local caps of [5, Lemmas 2.1 and 2.3] among its constraints, is infeasible. As in Razborov's flag algebras [16], the unknowns of the relaxation are the densities of the colourings of K_4 or K_5, the patterns, in a putative colouring of K_{61}. They are subject to linear constraints from the caps, such as $|N_i(v) \cap N_j(u)| \le \mathrm{R}(3,3) - 1 = 5$ for $u \in N_i(v)$ and $j \ne i$ (an edge of colour i or j inside this set would form a triangle of that colour with v or with u, so the set carries only the two other colours, both triangle-free), and to positive-semidefiniteness constraints. A certificate is a dual solution that proves the relaxation infeasible. For comparison, the six-vertex computation of Lidický and Pfender [13, Appendix A], whose relaxation has no caps, gives only $\mathrm{R}(3,3,3,3) \le 65$.

At $n = 61$ each colour-i degree is at most $\mathrm{R}(3,3,3) - 1 = 16$ [10], and the four colour degrees of a vertex sum to 60. So the *profile* of a vertex, its sorted quadruple of colour degrees, is one of

$$(12,16,16,16), \quad (13,15,16,16), \quad (14,14,16,16), \quad (14,15,15,16), \quad (15,15,15,15).$$

Both relaxations are numerically feasible. Floating-point solvers found an assignment of pattern densities meeting all their constraints. A case split by vertex profiles does not help either. The solvers found such an assignment even with the added requirement that every vertex has profile $(12, 16, 16, 16)$, so the case of any such split that allows this profile at every vertex is not excluded. The four-vertex relaxation was solved as a linear programme by HiGHS [11], with its semidefinite constraint imposed by eigenvector cuts until no eigenvalue was below -10^{-8}. The five-vertex relaxation was solved as a semidefinite programme by Clarabel [8] at tolerance 10^{-8}. So we obtained no such certificate from these relaxations, and no bound follows.

5 Verification, literature and limitations

Ancillary files. This paper and [5] have the same ancillary files, which are also in the directory `ancillary/` of the repository

`https://github.com/Lachy-Dauth/ramsey-neighbourhood-counting`.

They contain the programs and data, among them:

- `two_colour_r3k/` for Theorem 2.1, with the inputs from [7] and [2] and the certificates of both chains;
- `two_colour_r4k/` for Theorem 3.1, with the lists of deleted vertices, `build_power_sub.py`, which rebuilds the two graphs from them, the exact checkers `rcheck.c` (direct search) and `qpsym.c` (orbit argument), and `verify_all.sh`, which checks the SHA-256 hashes of the two graphs and runs the checkers;
- `flag5_r444/` and `flag5_r34k/`, which check the inputs $\mathrm{R}(4,4,4) \leq 198$ and $\mathrm{R}(3,4,4) \leq 73$ of Theorem 4.1 from [5].

The command `sh run_all.sh` runs the quick checks of both papers in a few minutes. The command `sh run_all.sh --full` adds the long ones, among them the checks of these two inputs and the direct search of the 369-vertex graph for independent 24-sets.

The checkers for this paper, in `two_colour_r3k/` and `two_colour_r4k/`, are written in Python and C with their standard libraries, and the C programs use POSIX threads. To run only these checks, use the commands in `two_colour_r3k/README.md` and the script `two_colour_r4k/verify_all.sh`. The script `run_all.sh` also needs a C++17 compiler, for the checks of [5], and uses NumPy, when it is installed, for an optional second checker in `flag5_r444/` and its negative controls [5, Section 10]. Negative controls are all rejected: a weakened input, a zeroed multiplier, a removed sequence certificate, a value raised by one, a claimed bound below the proved one such as $\mathrm{R}(3,24) \leq 199$, and a deleted vertex put back.

Literature. We searched arXiv and the web up to 23 October 2026 for earlier statements of our new bounds, and found none.

- For $\mathrm{R}(3,k)$ with $k \geq 24$ the survey lists no upper bounds, and the upper bounds of [7] and of Lesser's thesis [12] stop at $k = 16$ and $k = 21$. Backelin's unpublished manuscript (Section 2) may contain bounds of this kind.
- For $\mathrm{R}(4,21)$ the best previous bound we know of, 257, follows from $\mathrm{R}(4,20) \geq 252$ in the preprint [19], which we have not checked. Without it the bound is 242 [18].

- For $\mathrm{R}(4, 24)$ the survey lists nothing, and 343 is implicit in [14].
- For the four-colour numbers the previous bounds are those of [15] and [3] in Table 1.

Such a search cannot establish priority, and we would welcome pointers to any earlier source.

Limitations. For $11 \le k \le 23$ the bootstrap of Section 2 improves none of the bounds in [15]. At $k = 18$ it would give $\mathrm{R}(3, 18) \le 120$ if its own lower bound $e(3, 17, 102) \ge 731$ were one larger. The second row of Table 2 rests on the lower bounds of [2], whose proofs are largely unpublished.

The deletion method of Section 3 does not reach $\mathrm{R}(4, 23) \ge 320$. The four-colour bounds rest on $\mathrm{R}(3, 4, 4) \le 73$ and $\mathrm{R}(4, 4, 4) \le 198$ [5, Theorems 8.3 and 8.2], which are computer-assisted, and on $\mathrm{R}(3, 3, 3, 4) \le 149$, which [15, 6.1(a)] derives from $\mathrm{R}(3, 3, 4) = 30$ [4] and $\mathrm{R}(3, 3, 3, 3) \le 62$ [6], both computer-assisted.

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Much of the work reported here was carried out with two AI systems, used under the direction of the author: OpenAI's Codex, based on GPT-6, and Claude Opus 5.5, a large language model developed by Anthropic. They searched for the arguments and certificates, wrote the search programs for this paper (not included) and the checking programs in the ancillary files, and drafted the text. Every certificate and witness is checked in exact arithmetic by the programs in the ancillary files, which can be run without either AI system. The author takes full responsibility for the contents of this paper.

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